

★ PART 1 — Applicant Profile EDA

Focus: creditworthiness, risk, segmentation

1. Group applicants by credit_score range and show average annual_income.
2. Count applicants per employment_status.
3. Show applicants with debt_to_income > (SELECT AVG(debt_to_income)).
4. Rank applicants by credit_score using ROW_NUMBER().
5. Show average credit_score by housing_status.
6. Use a CTE to filter applicants under age 30, then show their average DTI.
7. Show applicants with delinquencies_past_2y > 1.
8. Show top 10 highest-income applicants.
9. Count applicants with bankruptcy_history = 1.
10. Use CASE WHEN to classify applicants into “Low/Medium/High Risk” based on DTI.

★ PART 2 — Application & Underwriting EDA

Focus: approvals, declines, credit lines, interest rates

11. Join applicants + applications to show cleaned approval status by individual credit score.
12. Show average assigned_credit_line by card_product.
13. Use a CTE to isolate declined applications, then count decline_reason frequency.
14. Show applicants whose requested_credit_line > assigned_credit_line .
15. Rank applicants by assigned_credit_line using RANK().
16. Show average interest_rate for approved applicants.
17. Use a subquery to find applicants with credit_score below the average of approved applicants.
18. Show approval_status grouped by employment_status.
19. Show applicants with DTI > 0.30 who were still approved.
20. Use CASE WHEN to label applications as “Over-Requested” or “Within Range.”

★ PART 3 — Transactions EDA

Focus: spending behavior, categories, card products

21. Show total spending per applicant_id.
22. Show total spending per category.
23. Use a CTE to isolate Retail transactions, then show average Retail spend.
24. Rank all transactions by amount using ROW_NUMBER().
25. Show average transaction amount by card_product.
26. Use a subquery to find applicants whose total spending > average total spending.
27. Show monthly spending totals.
28. Join applicants + transactions to show spending vs annual_income.
29. Use CASE WHEN to classify spending into “Low/Medium/High.”
30. Show total grocery spending for applicants with credit_score < 650.

★ PART 4 — Payments & Delinquency EDA

Focus: late payments, balances, minimum due, risk signals

31. Count applicants with at least one late payment.
32. Show average statement_balance by card_product.
33. Use a CTE to isolate late payments, then show average days_late.
34. Use LAG() to compare each applicant's payment_amount to their previous payment.
35. Join applicants + payments to analyze correlation between high DTI and late payments.